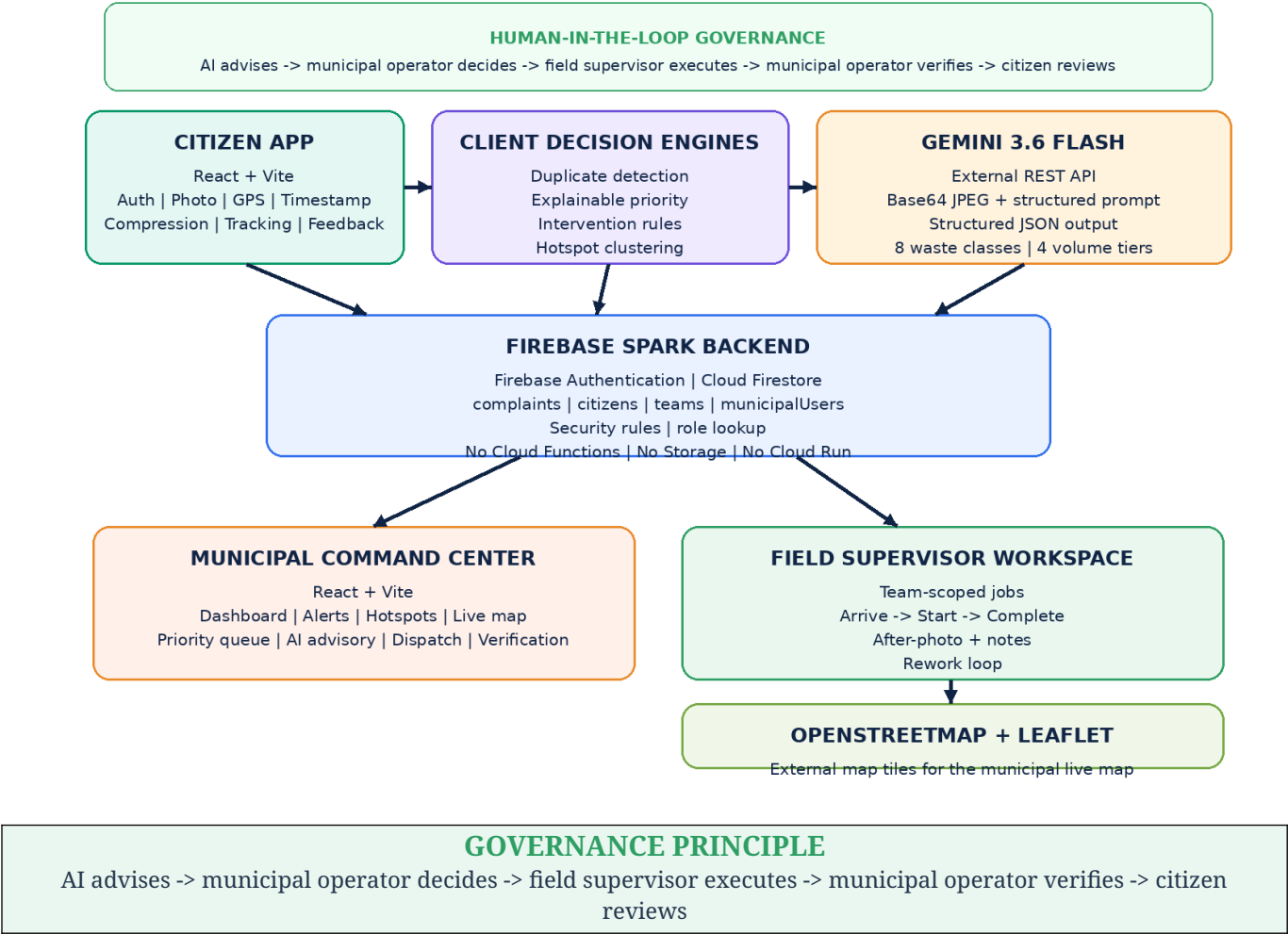


SwachhLens

Architecture & Data-Flow Technical Specification

TechNova: Igniting Brilliance (Season 3) - Problem Statement 2
Team TechTitans | GNIT Kolkata
Current prototype | Firebase Spark / no-cost architecture | Human-in-the-loop



1. System Modules & Functional Architecture

Module	Codebase / Tier	Core Function
Citizen Mobile PWA	citizen-app/src/ React + Vite	Camera/file capture, image compression, GPS, reporting, tracking, profile and feedback.
Client Decision Engines	citizen-app/src/services/ Browser JavaScript	dHash, duplicate detection, explainable priority, intervention rules, hotspot clustering.
External APIs	Gemini / Browser / OSM REST + HTTPS	Gemini 3.6 Flash, browser geolocation and external map tiles.
Firebase Infrastructure	Firebase Spark	Authentication, Cloud Firestore, indexes and security rules.
Municipal Command Center	portal/src/pages + components/ React + Vite	Dashboard, alerts, hotspots, live map, priority queue, dispatch and verification.
Field Supervisor Workspace	portal/src Supervisor*	Team-scoped jobs, Arrived -> Start -> Complete, after-photo evidence and rework.
Citizen Closure	citizen-app/src/components	Verified before/after review, 1-5 rating and reopen request.

2. External Services & Third-Party Dependencies

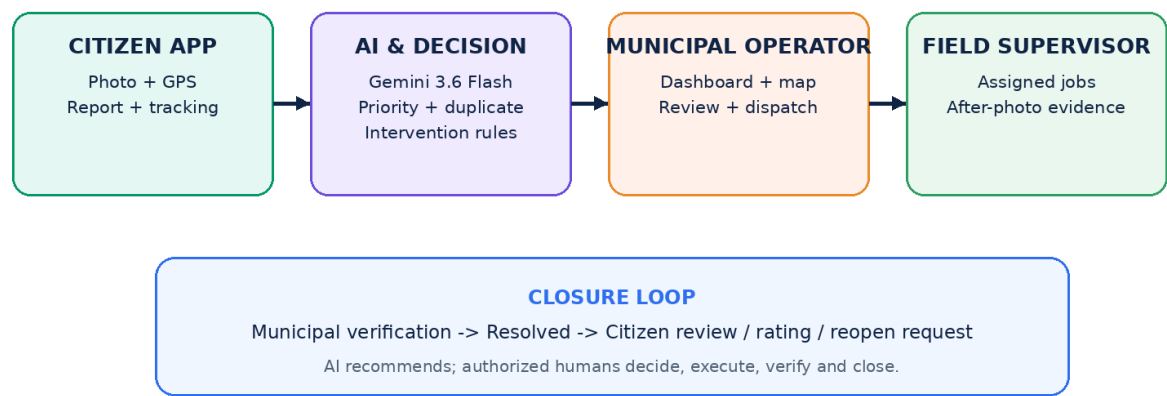
Service	Purpose	Boundary
Google Gemini 3.6 Flash	Multimodal waste-image analysis; returns structured JSON.	External REST API; image Base64 + prompt.
HTML5 Geolocation API	GPS latitude/longitude capture.	Browser API.
OpenStreetMap + Leaflet	Municipal live map and map tiles.	External tile service; network required.
Firebase Authentication	Anonymous guest and email/password identity.	Firebase Spark.
Cloud Firestore	Real-time data, persistence, queries and security rules.	Firebase Spark.

3. End-to-End Operational Data Flow

Step	Boundary / Interface	Data / Transition
1 Capture	Citizen device -> Canvas	Photo downscaled to <=800 px; JPEG Base64.
2 dHash	Canvas -> client hash engine	64-bit perceptual fingerprint.
3 Geolocation	Citizen app -> Geolocation API	Live lat/lng; demo fallback when permission denied.
4 AI inference	Citizen app -> Gemini API	Base64 JPEG + structured prompt -> JSON waste analysis.
5 Duplicate check	Decision engine -> Firestore query	Checks active candidates by category, time, GPS and image similarity.
6 Priority	Internal decision engine	Transparent 0-100 score with reason bullets.
7 Intervention	Internal rule engine	Maps waste attributes to recommended team/vehicle/crew/time.
8 Persistence	Citizen app -> Firestore	Writes complaint record and tracking ID.
9 Command center	Firestore -> portal	Real-time KPIs, alerts, map, hotspots and queue.
10 Human dispatch	Operator -> Dispatch UI	Accept recommendation or manual override.
11 Field execution	Supervisor -> Firestore	Arrive -> Start -> completion photo + notes.
12 Verification	Operator -> Verification UI	Before/after review -> Verify or Rework.
13 Citizen closure	Citizen app -> Firestore	Verified evidence -> rating, feedback or reopen request.

SwachhLens - Judge-Facing Architecture

Report -> Understand -> Prioritize -> Dispatch -> Execute -> Verify -> Close



4. Verified Algorithms & Decision Logic

PRIORITY SCORE

PriorityScore = min(100, round((V x 40) + (L x 30) + (F x 20) + (A x 10) + BioRiskBoost))

Factor	Verified rule
Volume	small 0.25 medium 0.50 large 0.75 very_large 1.00
Location	none 0.00 main_road 0.50 near school/hospital/waterbody 0.70 blocking drainage 1.00
Frequency	Nearby active reports within the implemented 50 m / 7-day window; capped at five.
Age	min(hours since report / 48, 1.0).
Bio-risk	+15 when bioWasteRisk is true.
Tiers	High >=70 Medium 40-69 Low <40.

5. Multi-Factor Duplicate Detection

A report is flagged rather than deleted when the implemented category/time/location checks identify a likely corroborating complaint. The current source material confirms a 50 m radius and 48-hour window. The architecture document intentionally omits the image-similarity percentage threshold because prior documentation used more than one threshold label; the detailed implementation value should remain in the master technical dossier after source-code confirmation.

6. Hotspot Intelligence

SwachhLens performs deterministic real-time spatial concentration clustering over active complaints using an 800 m radius and Haversine distance. It is current-state operational intelligence, not predictive forecasting.

7. Security & Role Boundaries

Role	Authentication	Operational scope
Anonymous Guest	Firebase Anonymous Auth	Report and view own session reports; no municipal portal access.
Registered Citizen	Email/password	Persistent profile, own reports and feedback.
Municipal Operator	Email/password + municipalUsers record	City-wide command center, dispatch and verification.
Field Supervisor	Email/password + field_supervisor role/teamId	Assigned-team jobs and completion evidence only.

8. Security Rules Verified in Final QA

Citizen/guest attempts to mutate protected complaint fields were rejected by Firestore.

Cross-citizen complaint/profile tampering was rejected.

Self-promotion into municipalUsers was rejected by the write lock.

Resolved-complaint citizen updates are restricted to the feedback field.

Hard document deletion is disabled by rules.

9. Technical Stack

Layer	Technology	Role
Frontend	React 19 + Vite 6	Citizen and municipal interfaces.
Routing	React Router 7	Client-side routing and role-aware views.
Data/Auth	Firebase Authentication + Cloud Firestore	Identity, persistence, real-time sync and rules.
AI	Google Gemini 3.6 Flash	Multimodal waste-image analysis.
GIS	Leaflet + React-Leaflet + OpenStreetMap	Live operational map.
Image engine	HTML5 Canvas API	Compression and perceptual hashing.
Icons	Lucide React	Operational interface iconography.

10. Prototype Boundary

This is a working hackathon prototype. It is not publicly deployed and is not adopted by a government body. The Spark-tier design intentionally avoids Cloud Storage, Cloud Functions and Cloud Run; images are compressed and stored as Base64 in Firestore. Production evolution would move AI inference server-side, use object storage for images, strengthen staff RBAC, and add explicit data retention/deletion controls.

Prepared for TechNova: Igniting Brilliance (Season 3) - Problem Statement 2 | Team TechTitans | GNIT Kolkata